Methylome analysis using MeDIP-seq with low DNA concentrations

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DNA methylation is an epigenetic mark that has a crucial role in many biological processes. To understand the functional consequences of DNA methylation on phenotypic plasticity, a genome-wide analysis should be embraced. This in turn requires a technique that balances accuracy, genome coverage, resolution and cost, yet is low in DNA input in order to minimize the drain on precious samples. Methylated DNA immunoprecipitation-sequencing (MeDIP-seq) fulfils these criteria, combining MeDIP with massively parallel DNA sequencing. Here we report an improved protocol using 100-fold less genomic DNA than that commonly used. We show comparable results for specificity (>97%) and enrichment (>100-fold) over a wide range of DNA concentrations (5,000–50 ng) and demonstrate the utility of the protocol for the generation of methylomes from rare bone marrow cells using 160–300 ng of starting DNA. The protocol described here, i.e., DNA extraction to generation of MeDIP-seq library, can be completed within 3–5 d.

INTRODUCTION

DNA methylation refers to the postreplicative maintenance or *de novo* addition of a methyl group to the carbon-5 position of the cytosine pyrimidine ring by DNA methyltransferases to form 5-methylcytosine (mC)[1]. In mammalian cells, the context in which DNA methylation occurs is dichotomous: CG (mCG) and non-CG (mCHG or mCHH, referred to henceforth as 'mC'). mCG accounts for approximately 60–80% of total CG dinucleotides in both stem and differentiated cell types[2], and given the palindromic nature of CG dinucleotides in double-stranded DNA, both strands are methylated. mCG on a single strand—so-called hemi-methylation—also occurs, but probably only as a temporary state between cell division and the enzymatic action of the maintenance DNA methyltransferase, DNMT1.

In contrast, mC methylation is asymmetrical with regard to strandedness and is only present in significant quantities in stem cells. The functional significance of mC methylation remains unclear, but it may affect accessibility of certain transcription factor–binding sites and therefore determine the activity of stem cell–specific transcriptional networks[3]. In addition, variant forms of mC (and mCG) were recently discovered, including 5-hydroxymethylcytosine (hmC)[4,5], 5-formylcytosine and 5-carboxylcytosine[6,7]. Although structurally similar to mC, hmC is much less abundant and is generated by the TET family of dioxygenase enzymes. Although its exact function remains unclear, it may form part of a mechanism that earmarks certain regions of the genome for constitutive hypomethylation.

Conversely, mCG methylation has been extensively studied, and its role in gene regulation is well understood. DNA methylation at CG-rich sites, so-called CpG islands that are commonly found in gene promoter regions, is associated with transcriptional repression, whereas the loss of methylation in gene bodies and particularly in repeat sequences is associated with genome instability[8]. The importance of DNA methylation in carcinogenesis is widely appreciated; however, its role in other common diseases, cellular homeostasis, mammalian development and aging is only just beginning to be elucidated. The availability of methods[9,10] for probing whole-genome methylation ('methylomes') has improved markedly in the past few years and holds much promise to improve our understanding of how DNA methylation regulates aspects of organismal biogenesis and development.

Protocol development

MeDIP is a technology capable of targeting the vast majority of the methylome. It involves antibodies directed against mC or mCG to precipitate methylated DNA fragments. MeDIP is able to detect methylated cytosines in both mC and mCG contexts, and, with the appropriate antibody, also hmC (hMeDIP)[10–12] and possibly 5-formylcytosine and 5-carboxylcytosine as well (please note that given the currently limited characterization of anti-hmC antibodies, this paper will focus on MeDIP only).

Because antibodies used for MeDIP were raised in a way to yield equal specificity against mC and mCG, MeDIP offers a near-unbiased and hypothesis-free approach without *a priori* assumptions about which regions of the methylome might be targeted (see below). Combining MeDIP with next-generation sequencing (MeDIP-seq[13]) provides high-quality methylomes at typically 100- to 300-bp resolution (depending on chosen insert size) at costs comparable to other capture-based techniques[14]. Indeed, MeDIP-seq was used to generate the first methylome of any mammalian genome[13]. We routinely use MeDIP-seq as our method of choice for analyzing methylomes, but found the high DNA concentrations (5–20 μg have typically been reported) required to be restrictive for some applications. A MeDIP-chip protocol for 150 ng starting DNA concentration was recently published[15]; however, the majority of DNA loss in MeDIP-seq occurs during library preparation, which is not carried out for MeDIP-chip. To address this, we have optimized the original MeDIP-seq protocol to work with as little as 50 ng DNA, making MeDIP-seq applicable to a much wider range of samples and studies. We have successfully applied this protocol to generate methylomes of purified mouse bone marrow cells (BMCs) using 160–300 ng of genomic DNA (O.T., unpublished data).

Comparisons with other methods

Current sequencing-based methods for methylome analysis can be classified into two groups: bisulfite conversion–based and

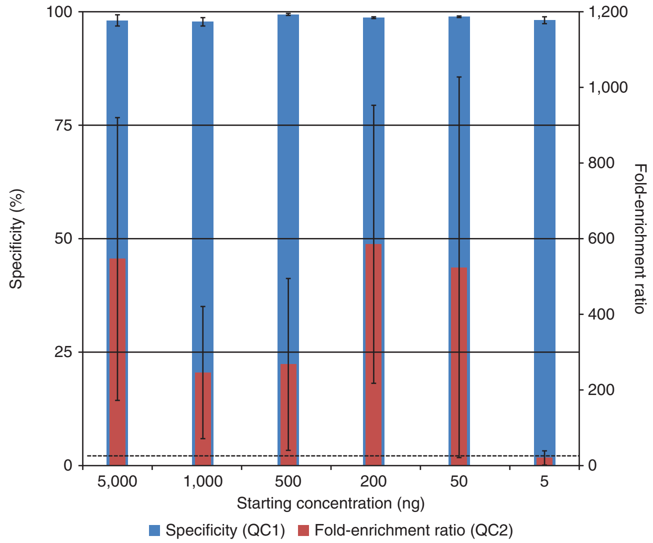


Figure 1 | Specificity and enrichment after QC1 and QC2 of test library–prepped MeDIP samples. The protocol was tested on four replicates of six starting DNA concentrations (5,000, 1,000, 500, 200, 50 and 5 ng). Constant antibody concentration and reaction volumes were used throughout. Spiked-in λ-DNA was assessed in QC1. Genomic regions of the fragmented sample DNA were assessed in QC2. The plot shows that the protocol can be successfully used with as little as 50 ng of starting DNA and possibly as little as 5 ng with further optimizations. The horizontal dotted line indicates a fold-enrichment ratio threshold of 25, which we deem the minimum value to pass samples on for sequencing. Error bars represent ± 1 s.e.m.

enrichment-based methods. Bisulfite conversion–based methods include whole-genome bisulfite sequencing (MethylC-seq or BS-seq)[16–18] and reduced representation bisulfite sequencing (RRBS)[19]. The latter category is also suitable for very low concentrations (5–100 ng) of genomic DNA. Despite their inability to distinguish between mC and hmC[20], bisulfite conversion–based methods are still considered to be the gold standard of DNA methylation analysis, as they permit quantification at single-base resolution. However, each variant has its drawbacks. For example, whole-genome bisulfite sequencing is currently prohibitively expensive for application on large sample sizes and by smaller research groups; RRBS only provides limited genome coverage (5–10%) and is CpG island and promoter region centric. Along with MeDIP-seq, the other main enrichment-based technologies are MBD-seq[21,22], which uses the methyl-binding proteins MBD2 and MBD3L1, and methylCap-seq[23], which uses the methyl-binding domain of MECP2 for methyl capture. Both are restricted to the analysis of mCG only and no whole-genome protocols have been reported for either of them for use with low (< 1,000 ng) concentrations of genomic DNA.

Limitations of the approach

One limitation of MeDIP-seq concerns genomic resolution. Our MeDIP-seq experiments provide a 150- to 200-bp resolution of the methylome, concomitant with sequence inset size, and although single–base pair resolution is inherently desired, we feel that the resolution offered by MeDIP-seq offsets issues of coverage and cost associated with bisulfite technologies. We also consider 150–200 bp to be a suitable resolution for most applications, as DNA methylation at adjacent CpGs is correlated for up to approximately 1 kb (ref. 24).

Another limitation of MeDIP-seq is that methylated DNA recovery by the antibody is affected by mC/mCG density, such that regions of very low (<1.5%) density may be underrepresented or even interpreted as unmethylated. Furthermore, MeDIP enriches only for methylated portions of the genome, and unmethylated portions can only be inferred by an absence of reads. The confidence placed on this inference depends highly on sequencing depth. To overcome this limitation, a hybrid method that combines MeDIP-seq with MRE-seq, a restriction-based method that enriches for unmethylated fractions of the genome, has been suggested[25].

Application of the method

In addition to being a cost-effective method for analyzing all currently known forms of mammalian DNA methylation, this optimized MeDIP-seq protocol has the advantage of requiring low-input DNA, which marks a considerable improvement from the original protocols[13,26], which require at least 5,000 ng DNA. This makes MeDIP-seq suitable for studies involving minute clinical samples, microdissected tissues and rare cell types.

TABLE 1 | Quality control and coverage data for the seven samples that were subjected to MeDIP-seq and used to create **Figure 1**.

Starting DNA concentration	Sample	Type	Specificity (QC1), %	CpG coverage (1×) at 17 million reads (%)
5 µg	Human normal PBMC DNA	36 bp PE	98.8	48.2
5 µg	Human normal PBMC DNA	36 bp PE	98.4*	47.2
600 ng	Dog tumor DNA_1	36 bp PE	98.5	56.4
600 ng	Dog tumor DNA_2	36 bp PE	97.6	56.6
300 ng	WT(a) mouse BM DNA_2	36 bp PE	99.9*	51.6
300 ng	WT(b) mouse BM DNA_1	36 bp PE	99.8	50.0
160 ng	WT(a) mouse BM DNA_1	36 bp PE	99.8*	50.6

BM, bone marrow; PBMC, peripheral blood mononuclear cell; WT, wild type.
On the basis of QC1, the specificity is >97% for all concentrations tested. Values marked by asterisks (*) depict samples where QC1 was conducted on the genomic DNA rather than the spiked-in λ-DNA. To determine whether MeDIP-seq read coverage is comparable for different DNA starting concentrations, 17 million random reads from each sample were analyzed for 1× CpG coverage and shown to be comparable within 10%.

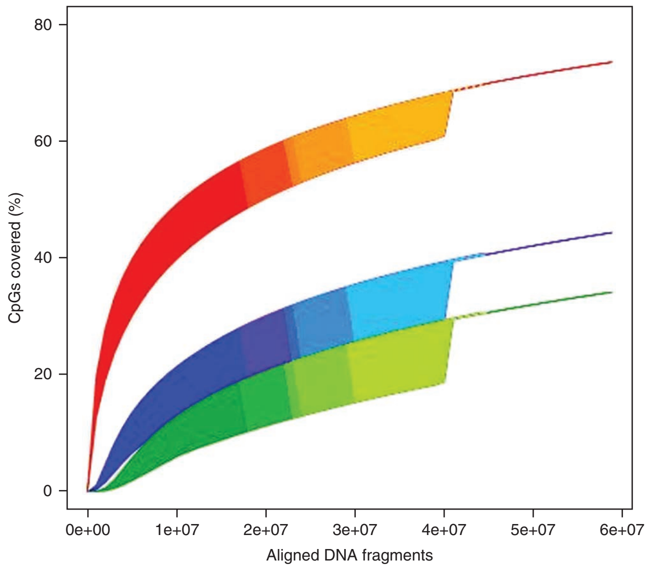
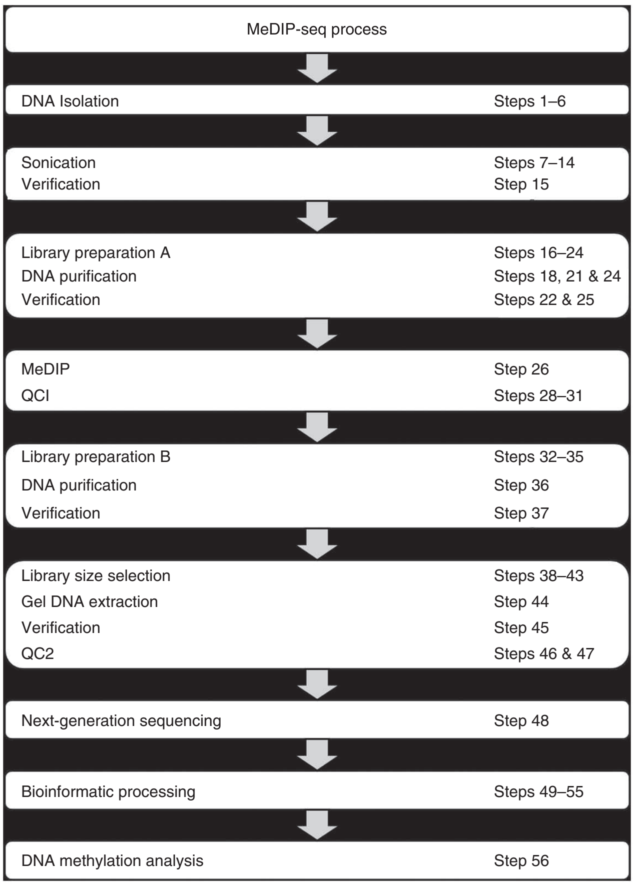


Figure 2 | Percentage of genomic CpGs at various sequencing depths as a function of MeDIP-seq read count. The plot shows data from seven samples derived from starting concentrations of 5,000, 600, 300 and 160 ng (two replicates each, except for 160 ng). The three shaded curves depict the number of times a CpG is covered by either a single read (1×; red), five reads (5×; blue) or ten reads (10×; green); the color gradient and banding of the lines indicate the number of samples included in the analysis (darkest, n = 7; lightest, n = 1), whereas the line's thickness indicates the observed upper and lower data range. All samples—regardless of starting concentration—follow a similar trend of CpGs covered, with the effect that more reads cover progressively fewer unique CpGs. The plot suggests that ~60 million reads (after filtering) will interrogate the majority of mCG in the genome.

The protocol described here has been optimized down to 50 ng of genomic input DNA, but has been tested on as little as 5 ng. In all, we have tested four replicates of six concentrations—5,000, 1,000, 500, 200, 50 and 5 ng of DNA—from human peripheral blood mononuclear cells (O.T., G.A.W., T.M., S.B. and L.M.B., unpublished data). **Figure 1** summarizes the results, showing excellent levels of specificity and enrichment achieved from starting DNA concentrations as low as 50 ng (O.T., G.A.W., T.M., S.B. and L.M.B., unpublished data), which are well above the empirical threshold determined previously for successful MeDIP-seq[26]. Although specificity is still excellent for 5 ng of starting DNA (as quantified by spiked-in DNA), enrichment is insufficient for cost-effective MeDIP-seq without further optimization. Therefore, we do not recommend using less than 50 ng of genomic DNA. For ease of use and potential high-throughput application, we chose to optimize a single protocol (e.g., with fixed antibody concentration, volume and so on) rather than optimizing the conditions for each DNA concentration tested. This is reflected in the variable level of enrichment and is likely to be caused by the varying amount of DNA competing for antibody binding.

We have applied this protocol to rare cells isolated from bone marrow starting with as low as 160 ng of DNA, and we have obtained high-quality MeDIP-seq data (O.T., unpublished data) comparable to those obtained from fresh blood by using 5,000 ng DNA[24]. Furthermore, the protocol was applied to four DNA concentrations from different sources in two replicates each for 5,000 ng (human DNA; M. Lechner, unpublished data), 600 ng (dog DNA; A. Fassati, unpublished data) and 300 ng, and a single replicate for 160 ng (mouse DNA; O.T., unpublished data); **Table 1** and **Figure 2** summarize the results, showing comparable specificity and coverage, thereby revealing that the percentage of CpGs covered with a fixed number of reads is comparable regardless of starting DNA concentration, and that CpG coverage can easily be enhanced by increasing the number of reads per sample (higher sequencing depth).

In our laboratory, the entire workflow (**Fig. 3**), from genomic DNA isolation to MeDIP-seq library, can be completed in 3 d; however, inexperienced users might prefer to spread the protocol over 5 d. The time required for the actual sequencing depends on the model of the sequencer (for an Illumina GAIIx, it takes ~7 d for a paired-end run of 36 bases), whereas the corresponding bioinformatic processing takes ~10 h.



Experimental design

Sample preparation from rare cell types. To analyze DNA from mouse BMCs, we purified ~75,000 BMCs by fluorescence-activated cell sorting, and pelleted, lysed and stored them at −70 °C for DNA/RNA extraction. For the rare BMCs, dual DNA and RNA extraction from the same cells can be carried out using the Qiagen AllPrep Micro kit according to the manufacturer's protocol. This kit also

Figure 3 | Protocol workflow from DNA extraction to sequencing. Each box represents the steps in the nano-MeDIP-seq protocol. The protocol must be completed in the above order as indicated by the direction of arrows above each box. 'Verification' refers to DNA analysis on an Agilent Bioanalyzer.

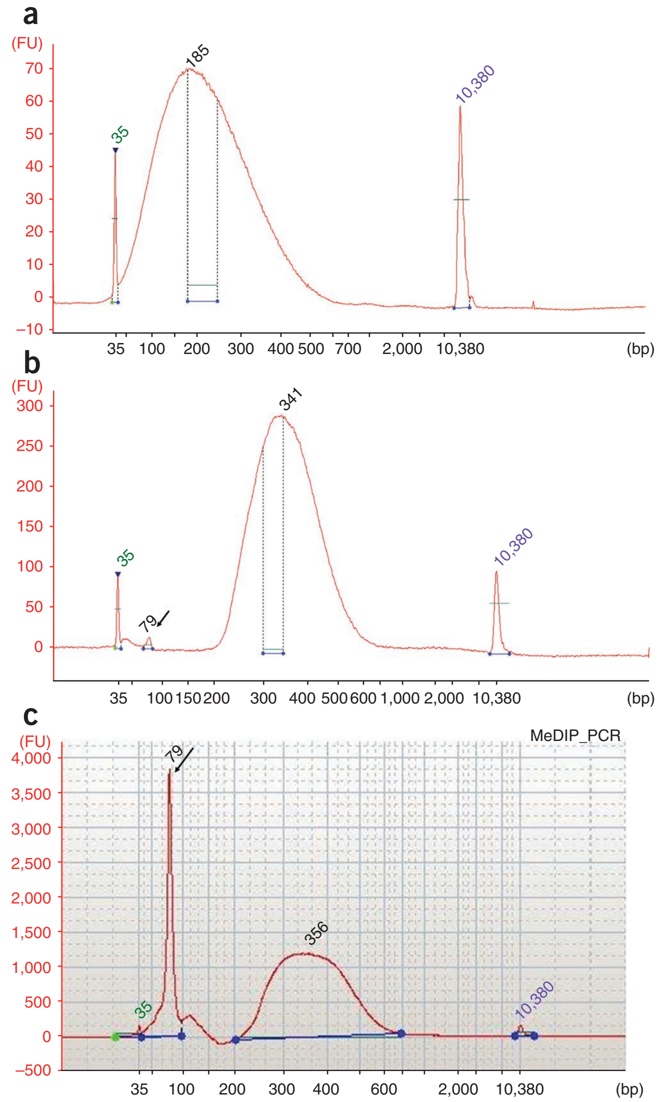


Figure 4 | Electropherograms of High-Sensitivity chips on an Agilent Bioanalyzer 2100. (**a**) The desired pattern of fragmentation following sonication. The peak occurs between 150 and 250 bp, which comprises approximately 25–30% of all fragments, which greatly enhances the yield of a MeDIP-seq library (typically 180–230 bp; dotted lines) following gel excision. (**b**) The fragment distribution following amPCR amplification of MeDIP DNA; the optimized PCR conditions permit retention of the original fragmentation pattern. We expect to see a shift in the peak by approximately 130–160 bp compared with fragmented DNA because of the incorporation of amPCR primer sequences. (**b**,**c**) The sample shown in **b** was purified using Ampure XP beads, which was more efficient at excluding primer dimers (arrowed) than Qiagen purification (**c**).

has the advantage that proteins can be simultaneously extracted. By performing a dual extraction, expression analysis can be performed on the same biological sample, which would otherwise be impossible for rare cell types. In addition, a dual extraction helps to reduce the number of animals needed to complete experiments. To minimize nucleic acid loss, we elute, store and perform all reactions in Eppendorf LoBind microcentrifuge tubes.

DNA fragmentation. To minimize DNA loss during fragmentation, we fragment DNA in LoBind tubes, as DNA can stick to the tube walls during this process. Our preference is to fragment by sonication with the Diagenode Bioruptor. After fragmentation, we assess the size range of the genomic DNA using an Agilent 2100 Bioanalyzer and high-sensitivity DNA chips. For our MeDIP-seq experiments, we chose to sequence library insert sizes of 180–230 bp; to maximize the quantity of DNA in this range, we aim to fragment the genomic DNA to 100–500 bp, with approximately 25–30% of the peak falling between 150 and 250 bp (**Fig. 4**). Experimenters may, however, vary the size of fragments generated depending on the required insert size. Nevertheless, it is essential that the majority of the fragmented DNA peak falls within the required insert size range, as this will eventually determine the level of the MeDIP-seq library recovery.

Library preparation A. To generate MeDIP-seq libraries, some adaptations to the generic manufacturer's (NEB) library preparation protocol are required. The main difference is that the generic library preparation is split into two phases, with the MeDIP procedure sandwiched between two library preparation steps. Following fragmentation, the first half of the library preparation ('A') consists of DNA end repair, 3′-dA tailing and adapter ligation. Most adapter ligation protocols require a strict DNA-to-adapter concentration ratio to ensure that adapter sequences are ligated to all DNA ends. As with the generic library preparation, we use a tenfold molar excess of adapter sequences to DNA. Given the stochastic nature of DNA clean up and sample-to-sample differences in fragmentation, we run the sample on an Agilent Bioanalyzer with High-Sensitivity DNA chips after 3′-dA tailing to determine sample molarity; we then halve this value (dA-tailed fragments are eluted in 25 µl; reaction volume of adapter ligation is 50 µl) and multiply by 10 to obtain the reaction concentration of adapter sequences. We prefer to perform paired-end sequencing over single-end sequencing for the reasons that this allows for better mapping of repeat regions, fragment length can be ascertained and that we retain more high-quality reads. The corresponding adapter and PCR primer sequences are detailed in **Table 2**.

MeDIP. After adapter ligation, we perform MeDIP. It is our preference to automate the MeDIP reaction with an SX-8G IP-Star and Auto-MeDIP kit reagents. Automation allows for the simultaneous processing of up to 16 samples and considerably reduces hands-on time, increases reproducibility and minimizes the risk of cross-sample contamination[26]. The use of automation is highly recommended, especially for low input DNA concentrations. However, a manual option may also be used via appropriate kits (such as MagMeDIP kit), although we have not validated our protocol with any manual kits.

We perform AutoMeDIP according to the manufacturer's protocol with minor modifications—the main one being the use of our custom-methylated and unmethylated controls. They consist of a cocktail of Enterobacteria phage lambda DNA (λ-DNA) fragments generated by PCR followed by *in vitro* methylation (where appropriate). Five fragments of varying CpG content are methylated and one is unmethylated, and there is no shared homology between them and mammalian genomes. The λ-DNA fragments were constructed as described in **Box 1**. All reagents except λ-DNA and sample DNA used in the Auto-MeDIP reaction are supplied as a kit from Diagenode.

TABLE 2 | Details of the oligos used in the protocol.

Oligo name	Genome	Sequence (5′→3′)	Length (bp)	CpG (%)	Amplicon size (bp)	Start	End
PE.Adapter.1.0	NA	ACACTCTTTCCCTACACGACGCTCTTCCGATC*T	33	NA	NA	NA	NA
PE.Adapter.2.0	NA	[Phos]GATCGGAAGAGCGGTTCAGCAGGAAT GCCGAG	32	NA	NA	NA	NA
PCR.primer.PE.1.0	NA	AATGATACGGCGACCACCGAGATCTACACTCTTT CCCTACACGACGCTCTTCCGATC*T	58	NA	NA	NA	NA
PCR.primer.PE.2.0	NA	CAAGCAGAAGACGGCATACGAGATCGGTCTCG GCATTCCTGCTGAACCGCTCTTCCGATC*T	61	NA	NA	NA	NA
1CpG_qPCR_F	λ-DNA	ACAAGTTGTTTGATCTTTGC	20	NA	145	24,004	24,148
1CpG_qPCR_R		CCTATGAGCAACGTGTTAG	19				
5CpG_qPCR_F		CACTTGAATCTGTGGTTCAT	20	NA	130	25,942	26,071
5CpG_qPCR_R		TAGAAAAAGACAACTCTGGC	20				
10CpG_qPCR_F		GAACTCACACACAACACCA	19	NA	125	33,662	33,786
10CpG_qPCR_R		ACTCTGAATACCGACTCAAT	20				
15CpG_qPCR_F		TATCACTGTTGATTCTCGC	19	NA	121	28,669	28,789
15CpG_qPCR_R		GGTAAAGAGTTTGGATTAGG	20				
20CpG_L_qPCR_F		GGTGAACTTCCGATAGTG	18	NA	108	24,202	24,309
20CpG_L_qPCR_R		CAGTCATAGATGGTCGGT	18				
20CpG_S_qPCR_F		GTTAGAGCCTGCATAACG	18	NA	109	44,747	44,855
20CpG_S_qPCR_R		GAAAGAGCACTGGCTAAC	18				
1CpG_F		GAGGTGATAAAATTAACTGC	20	0.5	197	23,967	24,163
1CpG_R		GGCTCTACCATATCTCCTA	19				
5CpG_F		CATGTCCAGAGCTCATTC	18	4.0	270	25,869	26,138
5CpG_R		GTTTAAAATCACTAGGCGA	19				
10CpG_F		CTGACCATTTCCATCATTC	19	5.6	360	33,638	33,997
10CpG_R		GTAACTAAACAGGAGCCG	18				
15CpG_F		ATGTATCCATTGAGCATTGCC	21	6.4	462	28,488	28,949
15CpG_R		CACGAATCAGCGGTAAAGGT	20				
20CpG_L_F		GAGATATGGTAGAGCCGCAGA	21	8.0	275	24,148	24,643
20CpG_L_R		TTTCAGCAGCTACAGTCAGAATTT	24				
20CpG_S_F		CGATGGGTTAATTCGCTCGTTGTGG	25	14.6	403	44,589	44,863
20CpG_S_R		GCACAACGGAAAGAGCACTG	20				
MM_meth_qPCR_F	Mouse	CATGGCCCACAAAGTAATAAAA	22	16.0	93	Chr1:19,718,938	Chr1:19,719,030
MM_meth_qPCR_R		AACGACTTACAACGAGCTCAAA	22				
MM_unme_qPCR_F		GGCTAGAACTGACCAGACAGAC	22	16.0	86	Chr4:16,751,777	Chr4:16,751,862

(continued)

TABLE 2 | Details of the oligos used in the protocol (continued).

Oligo name	Genome	Sequence (5′→3′)	Length (bp)	CpG (%)	Amplicon size (bp)	Start	End
MM_unme_qPCR_R		ATCTGTAGCCAATCCTAGAGCA	22				
HS_meth_qPCR_F	Human	GGGAATATAAGGAGCGCACA	18	10.0	114	Chr22:29,607,176	Chr22:29,607,289
HS_meth_qPCR_R		TCGGTTAAAACGGTCAGGTC	18				
HS_unme_qPCR_F		CGAGGCGTGAGTTATTCCTG	20	10.0	110	Ch6:4,011,057	Chr6:4,011,166
HS_unme_qPCR_R		CTCTTGTGGCTGAGCTCCTT	20				

CpG% is defined as 2× number CpGs / fragment length (bp). For fragments derived from the fragmented sample DNA, CpG% was estimated by simulation. Enterobacteria phage lambda genomic Start and Stop coordinates were derived from accession number NC_001416; human genomic start and stop coordinates were derived from reference assembly GRCh37.p2; mouse genomic start and stop coordinates were derived from reference assembly MGSCv37. Asterisk (*) indicates a phosphothiolate modification and [Phos] indicates a 3′ phosphate group.
NA, not applicable.

Quality control (QC) 1. The aim of QC 1 is to assess the efficiency of the MeDIP reaction by quantifying the spiked-in λ-DNA (and, where known, genomic regions of known methylation status) using real-time quantitative PCR (qPCR). Because the λ-DNA fragments are of differing lengths, we use nested PCR primers (**Table 2**) to amplify regions of roughly equal size. From the qPCR results, we calculate the 'recovery' of individual λ-DNA fragments and use these data to derive a single 'specificity' metric to determine MeDIP

Box 1 | Creating methylated and unmethylated control regions

The following describes a one-off set of procedures to create a central laboratory resource of methylated and unmethylated control fragments that should be suitable for over a million MeDIP-seq assays.

1. For each control region, combine the following reagents in a 0.2-ml PCR tube (note that it is often more convenient to make a master mix excluding primers):

Reagent	Reaction, 1× (μl)
Reaction buffer IV (10×)	2.5
$MgCl_2$ (25 mM)	1.0
dNTP mix (20 mM)	1.0
Primer forward (**Table 2**)	1.25
Primer reverse (**Table 2**)	1.25
Ultrapure water	17.625
λ-DNA (0.0005 ng μl^{-1})	1.0
Taq DNA polymerase	0.125
Total volume per reaction	25.0

2. Pipette the mix and briefly centrifuge it in a microcentrifuge to consolidate the reactions.
3. Set up and run the following PCR program: incubate for 2 min at 94 °C; a total of 40 cycles of (20 s at 94 °C, 30 s at 53/65 °C, 60 s at 72 °C); and then 5 min at 72 °C.
4. Purify each amplicon on a Clean & Concentrator-5 column according to the manufacturer's instructions and elute in 10 μl ultrapure water.
5. Run the sample on an Agilent Bioanalyzer 2100 using DNA1000 chips and reagents according to the manufacturer's instructions. Verify DNA fragment size range and concentration using the instrument's software.
6. Divide each amplicon into two aliquots and label.
7. Set up one *in vitro* methylation reaction and one control reaction for each aliquot in a 0.2-ml PCR tube as follows:

Reagent	*In vitro* methylation reaction (μl)	Control reaction (μl)
Ultrapure water	10.0	11.0
NE buffer 2 (10×)	2.0	2.0
S-adenosylmethionine (1.6 mM)	2.0	2.0
Amplicon DNA	5.0	5.0
SssI methyltransferase (4 U μl^{-1})	1.0	—
Total volume/reaction	20.0	20.0

8. Incubate for 2 h at 37 °C, followed by 20 min at 65 °C.
9. Purify each reaction on separate Clean & Concentrate-5 columns, according to the manufacturer's instructions, and elute in 10 μl of ultrapure water.

(continued)

Box 1 | (continued)

10. Each amplicon contains at least one 5′-ACGT-3′ motif, which allows the methylation status of each region to be verified using the methylation-sensitive enzyme, *HpyCH4*IV. To do so, start by diluting 1 µl of each amplicon from each reaction 1 in 10 with ultrapure water.
11. Set up the following reaction for each diluted DNA:

Reagent	Reaction, 1× (µl)
NE buffer 1	2.5
*HpyCH4*IV (10 U µl^{-1})	1.0
Diluted DNA	10.0
Ultrapure water	11.5
Total volume per reaction	25.0

12. Incubate for 1 h at 37 °C and then for 20 min at 65 °C.
13. Purify each sample on a separate Clean & Concentrator-5 column, according to the manufacturer's instructions, and elute in 6 µl of ultrapure water.
14. Run the sample on an Agilent Bioanalyzer 2100 with DNA1000 chips and reagents according to the manufacturer's instructions. Validate DNA fragment size range and concentration by using the instrument's software.
▲ CRITICAL STEP *In vitro* methylated fragments will not be digested by *HpyCH4*IV and will result in intact, full-length fragments that correspond to the length of the original amplicon; unmethylated fragments will be digested at the restriction site and show multiple fragment lengths.
15. Create 1 nM working dilutions of methylated and unmethylated controls on the basis of Bioanalyzer readings.
16. Use the 1 nM dilutions to create an equimolar cocktail of control fragments consisting of methylated (0.5, 3.7, 5.6, 6.5 and 10% CpG density) and unmethylated (14.5% CpG density) fragments.
17. Store stock solutions and working dilutions (e.g., 10 µM) of *in vitro* methylated and unmethylated fragments indefinitely at −20 °C.
18. Run the sample on an Agilent Bioanalyzer 2100 with DNA High-Sensitivity chips and reagents according to the manufacturer's instructions. Validate DNA fragment size range and concentration by using the instrument's software.
▲ CRITICAL STEP The molarity of each fragment in the cocktail is expected to be 166 pM; otherwise, correct accordingly by adding more of the required fragment(s) to normalize concentrations and recheck on a Bioanalyzer 2100 with High-Sensitivity chips.
19. Dilute the cocktail in 1× TE buffer to create a 100 pM (each) stock cocktail.
20. Dilute aliquots of the 100 pM stock in 1× TE buffer to create working dilutions of 0.19 pM (or ~114,000 copies per µl).

For the vast majority of QC purposes, we find that qPCR of just two regions (one methylated (10% CpG density) and one unmethylated (14% CpG density)) is sufficient to evaluate MeDIP. However, the full range of fragments in the cocktail can be assayed if desired.

efficiency. A successful MeDIP experiment is one where there has been a specific (≥95%) recovery of methylated fragments over unmethylated fragments. Note that recovery estimations based on λ-DNA will only provide indirect information about the immunoprecipitation success/failure of the sample DNA. Although it is possible to determine the recovery of the sample DNA itself, such analysis requires prior knowledge of the methylation status of any regions assayed by qPCR.

Library preparation B. After successful validation of the MeDIP reaction, library preparation is continued. In contrast to the generic library preparation protocol, we perform a dual adapter-mediated PCR (amPCR) for both MeDIP and input fractions; this is followed by library size selection. Input DNA is an aliquot of the preimmunoprecipitated reaction mix that is used as a reference sample to validate the MeDIP reaction. We have found this workflow advantageous for a number of reasons. First, because λ-DNA is added after adapter ligation, these fragments are not enriched during amPCR, which ensures that only the sample DNA will be sequenced. Of course, researchers can opt to add λ-DNA to the sample DNA after fragmentation if detailed QC using sequencing is desired. Second, we have found that performing two separate amPCRs can boost the final PCR yield by up to 50%. Third, it allows a reduction in PCR cycles—too many of which can skew the size distribution generated toward shorter fragment lengths and increase the likelihood of clonal reads. **Figure 4b** illustrates that the optimized amPCR is characterized by a near-identical distribution of fragment lengths compared with the input DNA. Finally, PCR before library size selection ensures that we amplify a broad range of fragment sizes, which permits multiple library size selections from the same PCR reaction.

Library size selection. Next-generation sequencing with Solexa chemistry requires the use of uniform insert DNA sizes, and we prefer to generate insert sizes of 180–230 bp. This is achieved by running the entire volume of pooled and purified amPCR products on a low-melting-point (LMP) agarose gel and excising gel slices in the 300- to 350-bp size range. We excise libraries that are ~120 bp longer than the required insert size, as DNA fragment sizes increase because of adapter and sequencing primer incorporation. We also prepare contingency libraries by excising adjacent gel slices (250–300 bp and 350–400 bp). As MeDIP only enriches for methylated fragments, and our adapters are unmethylated, we do not expect to see any unligated adapters in MeDIP sequence libraries. Furthermore, we prepare sequencing libraries after amPCR, which ensures that any dimerized oligonucleotides are excluded during size selection. **Figure 5a** shows the expected post-amPCR smear with three libraries excised

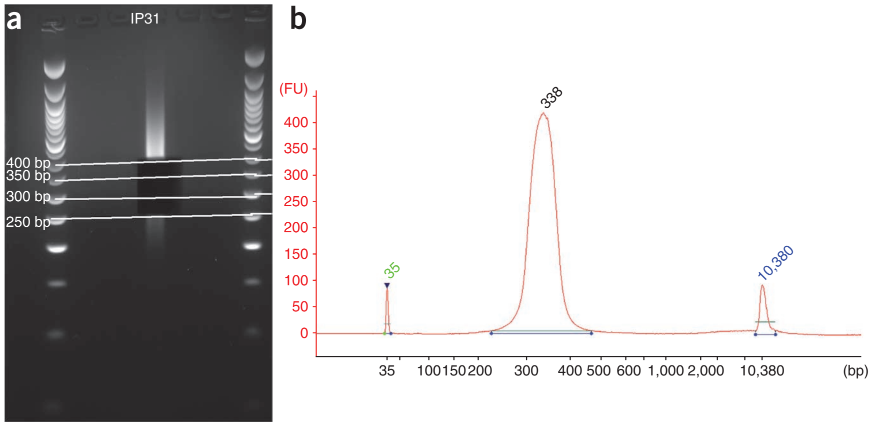


Figure 5 | Images from after gel excision. (**a**) We excise three bands (from between the white lines drawn on the gel image) from a MeDIP-seq experiment following amPCR. Although we intend to run only the 300- to 350-bp (180–230 bp insert size) library, 50-bp range contingency libraries are also excised in the event of irrevocable sample mix-ups or losses. (**b**) After purification, the concentrations of each MeDIP-seq library correlate with the original fragmentation pattern, with the highest concentration library corresponding to the peak after sonication.

from it. We routinely obtain ≥20 ng of purified sequencing libraries (depending on input DNA), which are completely free of contaminating oligonucleotides (**Fig. 5b**).

QC 2. QC2 involves the use of qPCR (as described in QC1) to confirm that the gel-extracted library remains enriched for methylated DNA while being depleted for unmethylated DNA. As the λ-DNA control fragments used in QC1 are not amplified by amPCR and therefore exist as negligible amounts in the library, methylated DNA enrichment is verified by quantifying selected regions of the genomic sample DNA whose methylation status is known on the basis of previously determined or published data[27].

DNA purification. It is essential that DNA be purified after each reaction. However, we find DNA loss during purification and gel extraction to be a limiting factor for MeDIP-seq with low-input DNA. To address this, we have compared two kits: Ampure XP beads and Qiagen columns. We recommend using Ampure XP beads for purification, as loss is minimal compared with cleanup using Qiagen kits, particularly for small quantities of DNA. Ampure XP beads are very efficient at removing DNA fragments of 100 bp or less; this is particularly important as excess adapter fragments and primer dimers interfere with PCR amplification of MeDIP and input samples (**Fig. 4c**). In addition, we have optimized the Qiagen gel extraction protocol to increase the yield of purified library DNA (**Box 2**).

Next-generation sequencing. After purification and quantification of the MeDIP-seq library on a Bioanalyzer with High-Sensitivity DNA chips, a minimum of 20 ng (2 ng μl^{-1}) of MeDIP-seq library is sequenced via the 36-bp paired-end module on the Illumina GAIIx. Other modules with single-end and/or longer read lengths can be used depending on context. The sequencing depth (e.g., number of lanes per sample) depends on a number of factors, including the size of genome and the purpose of the MeDIP-seq experiment. **Figure 2** illustrates the number of reads required to cover different proportions of CpGs at specific read depths, as well as how this can affect the appropriate number of lanes to use; at the time of writing, two lanes of sequencing on an Illumina GAIIx produces between 55 and 75 million raw reads, which should result in 45–60 million filtered and aligned, high-quality reads. With this data volume, ~70, 40 and 30% of all CpGs are covered at a minimum of 1×, 5× and 10× read depths, respectively. Bearing in mind that 60–80% of CpGs are methylated, a typical MeDIP-seq experiment might therefore interrogate the vast majority of methylated CpGs at 1× (**Fig. 2**). Obviously, higher coverage can be achieved by sequencing more lanes, and our protocol allows for additional lanes to be sequenced at a later date.

Alignment of reads and data analysis. For the analysis of MeDIP-seq data, we use two pipelines, BATMAN[13] and MeDUSA (methylated DNA utility for sequence analysis; G.A.W., unpublished data). MeDUSA (available upon request) is a publicly available computational pipeline script that brings together numerous freely available software packages to perform a complete analysis of MeDIP-seq data. MeDUSA controls several discrete stages of analysis, the first of which involves QC (FastQC; http://www.bioinformatics.bbsrc.ac.uk/projects/fastqc/), sequence alignment (BWA[28]) and filtering (SAMtools[29]). FastQC uses the Picard (http://picard.sourceforge.net/) suite of utilities and generates graphical representations of numerous quality metrics. The top of the FastQC HTML report shows a summary of the metrics that were run, and allows for a quick evaluation of whether the results seem normal (green tick), slightly unusual (orange triangle) or highly irregular (red cross). This evaluation is based on expectations from a 'normal' genomic sample. Of particular importance are the 'Per sequence quality score' and the 'Per base sequence quality', as these allow evaluation of the overall quality of the run.

Raw sequence data in .fastq format are aligned to the reference genome using alignment software such as BWA, with default parameters, to generate a SAM format alignment file.

Several filtering steps are performed on the alignment output in order to remove likely errors. Initial filtering to remove those reads failing to map as a proper pair is performed using SAMtools. SAMtools is used to filter reads so that only those with matching aligned pairs (forward and reverse templates) are retained. Further filtering based on mapping score is also performed (read pair must contain read with mapping quality ≥10). Finally, to remove potential clonal reads arising from PCR artifacts, for each group of nonunique reads (i.e., reads aligned to the exact same start and stop position on the same chromosome) all but one read are discarded.

In addition to prealignment QC using FastQC, MeDIP-seq–specific QC can be performed on the filtered data by using the Bioconductor package MEDIPS[30]. MEDIPS can be used to ascertain the reproducibility and CpG coverage of samples through performing saturation and coverage analyses. MEDIPS can also be used to generate genome-wide methylome profiles in WIG format, which are ready for viewing in genome browsers such as UCSC Genome Browser and Ensembl.

Box 2 | Optimized Qiagen gel extraction protocol

Elution volume is 10 µl, so use MinElute columns. Preheat EB and buffer QG (Qiagen buffer) to 50 °C. Dissolve agarose in 2 ml LoBind tubes

1. Weigh the gel slice in a colorless tube. A maximum of 400 mg of agarose can be processed per column; however, it is advised that a gel slice of less than 300 mg be processed per tube.
2. Add 3 volumes of buffer QG (preheated to 50 °C) to 1 volume of gel and vortex to mix.
3. Incubate the QG-gel mix in a 50 °C rotating incubator for 5 min. Ensure that the gel is completely dissolved. Incubate at RT for longer if necessary.
4. Check that the color of the mixture is yellow, similar to the color of buffer QG without the DNA sample. If there is a color change, add 10 µl of 3 M sodium acetate (pH 5.0) and mix. This will lower the pH of the mixture and the mixture will turn yellow.
5. Add 1 gel volume of isopropanol to the sample and mix by inverting several times.
6. Transfer the sample onto a MinElute column and centrifuge at maximum speed (~16,000*g*) for 1 min to bind DNA. The maximum volume of the column reservoir is 800 µl. For samples exceeding this volume, simply load and centrifuge multiple times.
▲ CRITICAL STEP Retain the tube used to mix gel DNA and isopropanol.
7. Discard the flow-through.
8. Add 500 µl of buffer QG (preheated to 50 °C) to the now-empty tube from step 6.
9. Vortex to mix.
10. Transfer the 500 µl buffer QG (from steps 8 and 9) to the MinElute column from step 6. Allow it to stand for 1 min and centrifuge at maximum speed for 1 min.
11. Discard the flow-through.
12. By using a pipette, aspirate and discard any residual buffer QG from inside the MinElute column before washing.
13. To wash, add 750 µl of PE wash buffer (Qiagen buffer) to the clean-up column.
▲ CRITICAL STEP Remember to add ethanol to buffer PE before use.
14. Gently invert the column several times to wash it thoroughly.
15. Allow it to remain undisturbed for 5 min.
16. Centrifuge at maximum speed for 1 min.
17. Discard the flow-through and place the clean-up column back in the same tube. Centrifuge the column for an additional 2 min.
18. Visually inspect the insides of the column to ensure that there is no residual solution in the column. Aspirate and discard any residual solution with a pipette.
19. Place the clean-up column in a clean 1.5-ml LoBind microcentrifuge tube.
20. To elute DNA, add the 10 µl of EB (preheated to 50 °C) directly onto the column membrane.
21. Allow it to remain undisturbed for 20 min.
22. Centrifuge at maximum speed for 1 min.

MATERIALS

REAGENTS

- End repair module (NEB, cat. no. E6050L)
- A-tail module (NEB, cat. no. E6053L)
- Adapter ligation module (NEB, cat. no. E6056L)
- High-fidelity Phusion polymerase (5×; NEB, cat. no. M0530L)
- dNTP mix (NEB, cat. no. N0447L)
- Auto-MeDIP kit (Diagenode, cat. no. AF-Auto01-0016 (16 reactions) or AF-Auto01-0100 (100 reactions)) ▲ CRITICAL MeDIP can also be performed manually using, e.g., the MagMeDIP kit (Diagenode, cat. no. mc-magme-A10) or other kits/reagents. In this case, we advise taking care with respect to accurate and consistent timing of incubation steps, as well as performing lengthy and thorough bead washing steps.
- IPure kit (Diagenode, cat. no. AL-Auto01-0100) ▲ CRITICAL MeDIP can also be performed manually using, e.g., the MagMeDIP kit (Diagenode, cat. no. mc-magme-A10) or other kits/reagents. In this case, we advise taking care with respect to accurate and consistent timing of incubation steps, as well as performing lengthy and thorough bead washing steps.
- Sequencing adapter and primer oligos (Sigma; see **Table 2**)
- QC primers (Sigma; see **Table 2**)
- MinElute Gel extraction kit (Qiagen, cat. no. 28004)
- Lambda (λ)-DNA (NEB, cat. no. N3011S)
- SssI CpG methyltransferase (NEB, cat. no. M0226S)
- Agencourt Ampure XP (60 ml; Beckman Coulter, cat. no. A63881)
- Ethanol (Sigma-Aldrich, cat. no. E7023) ! CAUTION Ethanol is highly flammable; keep flammable liquids away from all sources of ignition.
- Isopropanol (Sigma-Aldrich, cat. no. I9516) ! CAUTION Isopropanol is highly flammable; keep flammable liquids away from all sources of ignition.
- Sodium acetate, pH 5.2 (3 M; Sigma, cat. no. S7899-100ML)
- NuSieve low melting point Agarose (Lonza, cat. no. 50084)
- Ladder (50 bp; NEB, cat. no. N3236L)
- Tris Acetate EDTA (TAE, 50×; Severn Biotech, cat. no. 206001-10)
- Ethidium bromide (Sigma, cat. no. E1510-10ML) ! CAUTION Ethidium bromide is a highly toxic carcinogen; wear gloves and lab coats when handling ethidium bromide.
- Loading dye (Fermentas, cat. no. R0631)
- AllPrep DNA/RNA Micro Kit (Qiagen, cat. no. 80284)
- Agilent DNA 100 kit (Agilent, cat. no. 5067-1504)
- Agilent High-Sensitivity kit (Agilent, cat. no. 5067-4626)
- SYBR qPCR master mix (2×; Eurogentec, cat. no. RT-sy2x-03-Woub)
- Ultrapure water (VWR, cat. no. 23595.328)
- β-Mercaptoethanol (β-ME; Sigma, cat. no. M7522-100ML) ! CAUTION β-ME is highly toxic. Wear protective clothing, including gloves, eye and face masks when handling it.
- HCl
- DNA Clean & Concentrator-5 columns (Zymo)
- RLT plus lysis buffer (Qiagen)

EQUIPMENT

- Microcentrifuge tubes (0.5 ml; Eppendorf, cat. no. 0030108.035)
- Microcentrifuge tubes (1.5 ml; Eppendorf, cat. no. 0030108.0510)

- Microcentrifuge tubes (2 ml; Eppendorf, cat. no. 0030108.078)
- PCR tubes with flat cap (0.2 ml; Starlab, cat. no. I1402-8108)
- PCR tubes (8 strip; Starlab, cat. no. I1402-3500)
- PCR tube caps (8 strip; Starlab, cat. no. I1400-0900)
- PCR tubes (12 strip; Starlab, cat. no. I1402-1200)
- PCR tube caps (12 strip; Starlab, cat. no. I1400-1290)
- IP-Star tips (Diagenode, cat. no. WC-001-1000)
- qPCR plate (96 well; Starlab, cat. no. I1402-9800)
- Clear Type D qPCR plate seal (Starlab, cat. no. E2796-9795)
- Benchtop mini centrifuge with 0.2-ml tube adapter (Jencons PLC, cat. no. C1301-230V)
- Pipettes
- Pipette filter tips
- Vortex
- Transilluminator
- G.box gel imager (Syngene)
- Microwave oven
- Magnetic separation rack
- Single-pouch Sterile 10A scalpel (Swann-Morton or equivalent)
- UV cross-linker (Hoefer or equivalent)
- Glass bottles (250 ml; VWR or equivalent)
- Thermocycler machine (96-well plate; MJ research or equivalent)
- AB 7300 real-time PCR system (Applied Biosystems or equivalent)
- Gel electrophoresis system
- Gel electrophoresis tanks
- Gel comb (3 mm; Owl Separation Systems or equivalent)
- Weighing scale
- Weighing boats
- NanoDrop spectrophotometer (Labtech or equivalent)
  ▲ CRITICAL Alternative methods to the NanoDrop can be used to quantify DNA; however, care must be taken to ensure accurate quantification.
- Incubator oven (Techne or equivalent)
- Bioruptor UCD-200 sonicator (Diagenode) ▲ CRITICAL We recommend using the Bioruptor sonicator for DNA fragmentation; however, other methods can be applied provided these can generate the specified DNA fragment range and peak.
- Refrigerated microcentrifuge (Eppendorf or an equivalent)
- Heat block
- Agilent Bioanalyzer (Agilent)
- Illumina GAIIx (Illumina)
- SX-8G IP-Star (Diagenode) ▲ CRITICAL We use the SX-8G IP-Star for our automated MeDIP-seq protocol, as it was specially designed and programmed for this purpose; however, generic liquid handlers can be programmed to carry out the MeDIP step as detailed in the MagMeDIP protocol (Diagenode).

REAGENT SETUP

Tris solution, 1 M Dissolve 121.1 g of Trizma base in 1,000 ml of ultrapure water. Store at room temperature (i.e., 20–25 °C) and use within 18 months.

Tris-HCl, 1 M (pH 7.8) Dissolve 121.1 g of Trizma base in 700 ml of ultrapure water. Add HCl (36.5–38.0%) and test the pH. When pH reaches 7.8, fill up to 1,000 ml with ultrapure water. Store at room temperature and use within 18 months.

NaCl, 5 M Dissolve 292.2 g of NaCl in 1,000 ml of ultrapure water. Store at room temperature and use within 2 years.

STE buffer, 1× Mix 100 µl (1 M stock) of Tris-HCl (pH 7.8), 20 µl (0.5 M stock) of EDTA (pH 8.0) and 100 µl (5 M stock) of NaCl in a total volume of 10 ml (fill up with ultrapure water). Store at room temperature and use within 6 months. Final concentrations: 10 mM Tris-HCl, 1 mM EDTA and 50 mM NaCl.

TE buffer, 1× Mix 1 ml (1 M stock) of Tris solution, 200 µl (0.5 M stock) of EDTA solution and 40 µl (36.5–38.0%) of HCl solution in a total volume of 100 ml (fill up with ultrapure water). Store at room temperature and use within 2 years. Final concentrations: 10 mM Tris-HCl, 1 mM EDTA (pH 7.8).

Cell lysis buffer To aid the simultaneous isolation of total RNA from cells, add 10 µl of β-ME to 1 ml of RLT plus lysis buffer (store at room temperature and use within 1 month). ! CAUTION β-ME is toxic; dispense in a fume cupboard and wear protective clothing.

Enterobacteria phage λ-DNA control fragments The construction of six λ-DNA fragments of varying CpG content is described in **Box 1**.

Paired-end (PE) sequencing adapters Adapter sequences (PE.Adapter.1.0 and PE.Adapter.2.0) are shown in **Table 2**. Spin down lyophilized adapter oligos in a chilled (4 °C) microcentrifuge at 200*g* for 5 min.
Resuspend each oligo in 1× STE buffer to 100 µM. Add equimolar quantities of each adapter into a 1.5-ml microcentrifuge tube. Divide into small (e.g., 200 µl) aliquots and incubate at 95 °C for 15 min, and then leave to cool down to room temperature (~1 h). Store aliquots at −20 °C until required. Final concentration: 50 µM.

Sequencing primers Adapter-mediated PCR primer sequences (PE.PCR.1.0 and PE.PCR.2.0) are shown in **Table 2**. Spin down primer oligos in a chilled (4 °C) microcentrifuge at 200*g* for 5 min. Resuspend each oligo in 1× TE buffer to create 100 µM stocks. Dilute an aliquot tenfold in 1× TE buffer to create 10 µM dilutions. Store dilutions in manageable volumes at −20 °C until required.

NEB ladder We use a 50-bp DNA ladder (NEB) with molecular-weight standards ranging from 50 to 700 bp in 50-bp size intervals. We find these intervals between 50 and 500 bp particularly useful, as they increase the accuracy of our gel excision during size selection.

QC primers For QC1, nested qPCR primers were designed for each λ-DNA fragment (**Table 2** and **Box 1**). qPCR primers for QC2 were designed to amplify regions of presumed methylation status in the reference genomes based on previous work[27]. ▲ CRITICAL Users are advised to validate methylation status of QC regions selected for their experiments. Primers for either methylated or unmethylated regions were matched such that each amplicon had similar CpG contents (**Table 2**). Store aliquots at −20 °C until required. Final concentration: 10 µM.

LMP agarose and TAE gels We prepare 2% (wt/vol) LMP TAE gels for use in library generation. Briefly, 2 g of LMP agarose gel is dissolved in 100 ml 1× TAE. The solution is swirled gently to dissolve the agarose into solution while avoiding the generation of bubbles. Agarose mix is heated in a microwave oven at 800 W for 2 min. Leave the gel to cool to ~35 °C before adding ethidium bromide to a final concentration of 0.5 µl ml^{-1}. Pour the gel slowly into a casting tray that has been placed on an even surface. Avoid bubbles when casting gels. ! CAUTION Glassware will be very hot. Always wear protective gloves when handling hot substances. ! CAUTION Ethidium bromide is a known mutagen. Always wear protective gear when handling reagents. ▲ CRITICAL Loosen or uncover glassware while heating gel to avoid spills and swirl briefly after 1 min to mix. ▲ CRITICAL Use 3-mm combs as this creates small wells. We have found that DNA is recovered more efficiently from smaller gel slices. ▲ CRITICAL Prepare gels with enough wells to accommodate sample and two flanking ladder lanes with a minimum of three empty lanes spacing to avoid bleed-through into adjacent wells. ▲ CRITICAL To avoid cross-contamination of samples, it is advisable to prepare separate gels for each sample. All glassware and plastics used for gel electrophoresis are UV treated to eliminate any contaminating genomic material.

PROCEDURE

DNA/RNA isolation ● TIMING 1–2 h

1| Pellet 75,000 or more cells for 5 min at 300*g*.

2| Remove the supernatant by aspiration with a pipette.
▲ CRITICAL STEP A pellet of 75,000 cells in PBS is invisible to the naked eye; thus, care should be taken to note the exact orientation of tubes in the centrifuge. Cell pellets are expected to fall on the side of the microcentrifuge tube closest to the centrifuge wall. Extra care should be taken when aspirating supernatant from samples.

3| Loosen the cell pellets by flicking the tube.

4| Resuspend the cells in 350 μl of lysis buffer (RLT Plus).

5| Homogenize the cell lysates by vortexing for a minimum of 1 min.
■ **PAUSE POINT** Homogenized cell lysates can be stored at −70 °C for later DNA extraction and subsequent MeDIP-seq analysis.

6| Isolate genomic DNA (and RNA) using AllPrep DNA/RNA spin columns and reagents supplied in the Qiagen AllPrep kit according to the manufacturer's instructions.
▲ **CRITICAL STEP** Elute in no more than 85 μl of elution buffer (EB).
■ **PAUSE POINT** Isolated genomic DNA is relatively stable and can be stored at −20 °C for up to 1 year.

DNA sonication on Diagenode bioruptor ● TIMING 2–3 h

7| Resuspend ~300 ng of DNA (from Step 6) in 85 μl of EB in a 0.5-ml LoBind tube.

8| Prechill the sonicator water bath by filling it with ice and leaving it for 15 min.

9| After 15 min, replace the ice in the water bath with ice-cold water.

10| Place the microcentrifuge tube containing DNA in the tube holder, making sure that the holder is balanced with the centrifuge.

11| Set the sonicator to high and sonicate for four to six 15-min cycles, each consisting of 30-s on/off periods.

12| Cool the sonicator for 5–10 min between each 15-min cycle by filling it with crushed ice.

13| Briefly centrifuge the DNA samples between each cycle.

14| Keep the DNA below 10 °C during and after sonication.
▲ **CRITICAL STEP** If sonication is being performed for more than 60 min, repeat Step 8.
■ **PAUSE POINT** Fragmented DNA can be stored at −20 °C for up to 1 month.

15| Run the sample on an Agilent Bioanalyzer 2100 using High-Sensitivity DNA chips and reagents according to the manufacturer's instructions; validate DNA fragment size range and concentration with the instrument's software.

Library preparation A ● TIMING ~7 h

16| *DNA end repair.* Mix the following on ice in a sterile PCR tube and incubate the mixture in a thermocycler at 20 °C for 1 h:

Reagent	Reaction, ×1 (μl)
NEBNext-end repair enzyme mix	5
NEBNext-end repair reaction buffer (10×)	10
Fragmented DNA from Step 14	85
Total volume	100

17| Place the reaction on ice after incubation and spin briefly in a mini centrifuge for ~10 s to collect condensate.

18| Purify the DNA sample with Ampure XP purification beads according to the manufacturer's instructions and elute in 37 μl of EB.

19| *dA-tailing*. Mix the following on ice in a sterile PCR tube and incubate the mixture in a thermal cycler for 1 h at 37 °C:

Reagent	Reaction, ×1 (μl)
Klenow fragment (3′→5′ exo⁻)	3
NEBNext dA-tailing reaction buffer (10×)	5
Sterile H_2O	5
Purified end-repaired, blunt-ended DNA from Step 18	37
Total volume	50

▲ **CRITICAL STEP** To prevent evaporation of samples, use a heated thermocycler lid (100 °C) during all incubations involving temperatures over 30 °C.

20| Pipette up and down to mix.

21| Purify the DNA sample with Ampure XP purification beads and elute in 26 μl of EB.

22| Run 1 μl of eluted sample on an Agilent Bioanalyzer 2100 using High-Sensitivity DNA chips and reagents according to the manufacturer's instructions; determine molarity of the full range of DNA fragments with the instrument's software.
▲ **CRITICAL STEP** Calculate the reaction concentration of the DNA adapters to be used in the adapter ligation reaction as described in Experimental design.

23| Mix the following on ice in a sterile PCR tube and incubate the mixture in a thermocycler at 18 °C for 2 h:

Reagent	Reaction, ×1 (μl)
Quick T4 DNA ligase	5
Quick ligation reaction buffer (5×)	10
DNA adapters	10
End-repaired, blunt, dA-tailed DNA from Step 21	25
Total volume	50

24| Purify the DNA sample with Ampure XP purification beads and elute in 40 μl of EB.

25| Run 1 μl of eluted sample on an Agilent Bioanalyzer 2100 using High-Sensitivity DNA chips and reagents according to the manufacturer's instructions; determine the concentration of the full range of DNA fragments.
▲ **CRITICAL STEP** If multiple samples are to be run, use concentration data to normalize sample concentrations to maintain similar stoichiometry during MeDIP.

MeDIP ● TIMING 23–25 h (includes overnight incubation)

26| Perform MeDIP on purified adapter-ligated DNA from Step 24, by using the Diagenode Auto-MeDIP and iPure kits and SX-8G IP-Star, according to the manufacturer's instructions; **Box 3** describes AutoMeDIP setup and procedure (see **Fig. 6**).
▲ **CRITICAL STEP** Alternative manual methods may used (e.g., MagMeDIP (Diagenode)) according to the manufacturer's instructions.
■ **PAUSE POINT** After MeDIP, DNA can be stored at −20 °C for up to 3 months and longer at −80 °C.

qPCR to test for recovery of spiked-in λ-DNA and specificity of the MeDIP reaction (QC1) ● TIMING ~4 h

27| Dilute 4.5 μl of MeDIP and input DNA (from Step 26) to 10 μl with ultrapure water.

Box 3 | Setting up AutoMeDIP and loading the IP-Star

Auto-MeDIP setup

1. Prepare 'Antibody (Ab) Dilution' as follows:

Reagent	Vol (μl)
Ab	1
Ultrapure water	15
Total volume	16

2. Prepare 'Antibody Mix' as follows:

Reagent	1 IP (μl)	2 IPs (μl)	4 IPs (μl)	6 IPs (μl)
Diluted Ab (from Step 1)	2.40	6.00	12.00	16.00
MagBuffer A (5×)	0.60	1.50	3.00	4.00
MagBuffer C	2.00	5.00	10.00	13.00
Total volume	5.00	12.50	25.00	33.00

3. Prepare 'Incubation Mix' for one IP (and input) as follows and

Reagent	Reaction, ×1 (μl)
MagBuffer A (5×)	24
MagBuffer B	6
Lambda spike cocktail (0.2 μM each; see REAGENT SETUP)	3
Purified adapter-ligated DNA	Variable
Ultrapure water	to 90 μl

4. Incubate at 100 °C for 10 min.
5. Snap-cool on ice for 10 min.
6. Mix by pipetting up and down and spin for 10 s on a mini centrifuge.

▲ CRITICAL STEP Auto-MeDIP is performed by using the Diagenode Auto-MeDIP and iPure kits and SX-8G IP-star, according to the manufacturer's instructions; however, the above six steps deviate from the manufacturer's instructions and represent optimized steps for Auto-MeDIP setup.

Load the IP-Star

The IP-Star (Diagenode) is capable of automating up to 16 MeDIPs per run. Each run is split into two parts with a manual intervention step in between. Each part requires separate plating-out of reagents. After plating out the reagents for part one (see below), the IP-Star washes the magnetic beads (to remove any preservatives), immunoselects and captures the methylated fraction, washes away unbound (unmethylated) DNA and elutes in a chaotropic buffer. The second part is the automated purification of the methylated fraction from the antibody and beads. The result is a pure methylated fraction ready for amPCR. The following explains how to plate out reagents for these processes:

1. Switch on the IP-Star approximately 10 min before loading to equilibrate Peltier heat/cool blocks to 4 °C.
2. Load the following reagents into a 12-strip tube (one strip tube per IP):
   - Well 1: Empty
   - Well 2: Empty
   - Well 3: Empty
   - Well 4: 50 μl EB (from iPure kit)
   - Well 5: 10 μl 1× MagBuffer A + 10 μl beads
   - Well 6: 50 μl 1× MagBuffer A
   - Well 7: 75 μl Denatured incubation mix + 20 μl 5× MagBuffer A + 5 μl antibody mix
   - Well 8: 100 μl MagWashBuffer1
   - Well 9: 100 μl MagWashBuffer1
   - Well 10: 100 μl MagWashBuffer1
   - Well 11: 100 μl MagWashBuffer2
   - Well 12: 50 μl EB (from iPure kit)
3. Load the 12-strip tube containing the IP reagents onto the Peltier block(s) (**Fig. 6**; black-boxed zone). MeDIP proceeds in the direction of the arrow; therefore, align well 1 of the 12-strip tube with the beginning of the arrow.

▲ CRITICAL STEP If you are running ≤8 IPs, double-click tip rack 1 (**Fig. 6**; red-boxed zone) until the graphical user interface (GUI) shows the presence of tips; if you are running >8 IPs, double-click tip rack 1 and tip rack 2 (**Fig. 6**; red-boxed zone).

▲ CRITICAL STEP If you are running ≤8 IPs, load the right-hand Peltier heat/cool block and load 8 IP-Star tips per row in corresponding wells of tip rack 1 following the arrow direction (**Fig. 6**; red-boxed zone). If you are running >8 IPs, load the remainder of samples on the left-hand Peltier heat/cool block and add an extra four IP-Star tips to tip rack 1 and four tips to tip rack 2.

▲ CRITICAL STEP When you are running >8 IPs, it is important to load tip rack 2 rather than other tip racks, as this is the default search position after tip rack 1 becomes empty; if no tips are found in tip rack 2, the IP-Star will stall until user intervention specifies where it can find available tips.

(continued)

Box 3 | (continued)

▲ CRITICAL STEP When you are running >8 IPs, it is best to balance the workload across both Peltier blocks and mirror the layout between blocks. If you are running an odd number of IPs, we recommend placing an empty 12-strip tube on the Peltier block with fewer samples in order to mirror the formation of the Peltier block with more samples.
▲ CRITICAL STEP Select 8IPs or 16IPs protocol for 8 and 16 MeDIP reactions, respectively.
4. Click 'Modify' to set IP incubation time and temperature.
▲ CRITICAL STEP IP incubation times are for each block. We recommend at least 7.5 h of incubation time (15 h is preferable) and an incubation temperature of 4 °C.
5. Press Start and OK the confirmation screen that appears. After the MeDIP is completed, up to 16 samples can be purified on the IP-Star.
6. Load the following reagents (all from the iPure kit) into a Nunc U96 plate:
Well 1: 150 µl Buffer C
Well 2: 100 µl of isopropanol, 15 µl of beads, 2 µl of carrier + 100 µl of MeDIP/input (if input is being purified, 7.5 µl of input DNA is diluted to 100 µl of iPure EB)
Well 3: Wash Buffer 1
Well 4: Wash Buffer 2
Well 5: Empty (pending 25 ml elution)
Well 6: Empty (pending 25 ml elution)
Well 7: Empty
Well 8: 100 µl of isopropanol, 15 µl of beads, 2 µl of carrier + 100 µl of MeDIP/input (if input is being purified, 7.5 µl of input DNA is diluted to 100 µl of iPure EB)
Well 9: Wash Buffer 1
Well 10: Wash Buffer 2
Well 11: Empty (pending 25 ml elution)
Well 12: Empty (pending 25 ml elution)
▲ CRITICAL STEP If ≤8 samples are being purified, only fill wells 1–4.
7. Place the 96-well plate in the 96-well plate holder (**Fig. 6**; blue-boxed zone).
8. Purification proceeds in the direction of the arrow; therefore, align well 1 of the plate with the beginning of the arrow.
9. Double-click tip rack 1 until the GUI shows the presence of tips.
10. If ≤8 samples are being purified, load four IP-Star tips into corresponding rows of tip rack 1; if >8 samples are being purified, load eight IP-Star tips into corresponding rows of tip rack 1.
11. If ≤8 samples are being purified, 25 µl of purified product is collected from wells five and six and pooled (giving 50 µl); if >8 samples are being purified, 25 µl of purified product is also collected from wells 11 and 12 and pooled (giving 50 µl).
12. Discard the reaction plates and proceed to QC1 (Step 27).

28| Prepare the following on ice in a qPCR plate. Perform each reaction in triplicate for methylated and unmethylated control regions:

Reagent	Reaction, ×1 (µl)
SYBR qPCR master mix (2×)	6.25
Primer pair (10 µM)	0.625
Water	4.375
MeDIP or input DNA	1.25
Total volume per reaction	12.5

29| Seal the plate with a type D plate seal and centrifuge briefly.

30| Set up and run the following program on the real-time PCR machine: 95 °C for 5 min; 40 cycles of 95 °C for 15 s and 60 °C for 1 min; melt curve.

31| Calculate recovery and specificity according to **Box 4**.
▲ CRITICAL STEP Proceed with the next steps only if the MeDIP fraction shows a recovery of ≤1% (unmethylated) and a specificity of ≥95%.
? TROUBLESHOOTING

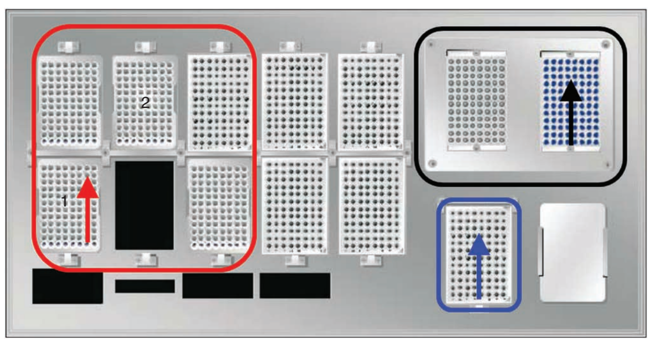


Figure 6 | Layout of the IP-Star bed. The red-boxed zone illustrates the location of tip holders (tip racks 1 and 2); the black-boxed zone shows the two 96-well Peltier blocks; and the blue-boxed zone demarcates the 96-well plate holders. The arrows demonstrate the direction of workflow of the IP-Star.

Box 4 | Calculations and interpretations of QC data by qPCR

QC1

From the qPCR results, recovery and specificity are calculated to evaluate MeDIP efficiency. Recovery, which is a measure of the MeDIP efficiency and is indexed as percentage input, is calculated using the cycle threshold (C_t) of MeDIP and input fractions from the qPCR reaction. As only 10% of the DNA used in MeDIP is used for input, the C_t value obtained for the input is adjusted before calculating recovery, by using the following formula:

$$\text{Adjusted Input Ct} = \text{Input Ct} - \log[2\text{AE}(10)]$$

Where AE is the amplification efficiency expressed as a number (e.g., 100% AE = 1) and 10 represents the dilution factor of input to MeDIP (i.e., 10×).

Recovery is then calculated as follows:

$$\text{Recovery}(\%) = 2\text{AE}^{(\text{Adjusted Input Ct} - \text{MeDIP Ct})} \times 100$$

Specificity of MeDIP, indexed as the ratio of methylated DNA recovery to unmethylated DNA is calculated as:

$$\text{Specificity} = 1 - \left\{\frac{\text{unmeth recovery}}{\text{meth recovery}}\right\}$$

We consider MeDIP to be successful when specificity is ≥95% and that this has been derived from unmethylated recovery of <1%.

QC2

The aim of QC2 is to verify the library insert remains enriched for methylated fragments and depleted for unmethylated fragments following amPCR and gel excision. In contrast to QC1, QC2 involves the qPCR of fragments from sample DNA. This is because the λ-DNA fragments are not capped with adapter sequences and not amplified during amPCR. By using data from previous studies or those that have been validated in-house[27] (for humans), primers have been designed for two regions that are likely to be differentially methylated in humans and mouse; these are presented in **Table 2**. After purifying the excised fragment, normalize the concentrations of MeDIP and input on the basis of the Bioanalyzer readings and set up triplicate qPCRs for the two genomic regions using the reaction mix and thermal cycling conditions described above (see PROCEDURE Step 38). Calculate the fold-enrichment ratio for methylated versus unmethylated regions:

$$\text{Fold Enrichment Ratio} = \frac{2\text{AE}^{(\text{Ct}_{\text{input\_meth}} - \text{Ct}_{\text{MeDIP\_meth}})}}{2\text{AE}^{(\text{Ct}_{\text{input\_unmeth}} - \text{Ct}_{\text{MeDIP\_unmeth}})}}$$

In the presence of equally efficient primers, the fold-enrichment ratio can be calculated without input DNA (i.e., $2\text{AE}^{(\text{Ct}_{\text{MeDIP\_unmeth}} - \text{Ct}_{\text{MeDIP\_meth}})}$); however, we recommend running each primer on at least one genomic DNA a priori to verify equal efficiency. We recommend passing libraries on for sequencing if the fold-enrichment ratio exceeds 25, i.e., the methylated fragments tested are 25-fold more enriched than the unmethylated fragments. However, we have routinely observed that fold-enrichment ratios of >100 and higher scores are better.

Library preparation B ● TIMING ~3 h

32| Split MeDIP and input fractions into two parts and perform the following reaction for each half.

33| On ice, mix the following in a sterile PCR tube. Pipette up and down to mix:

Reagent	Reaction, ×1 (μl)
PCR.PE.1.0 (10 μM stock)	5
PCR.PE.2.0 (10 μM stock)	5
MeDIP or input DNA	25
Phusion high-fidelity polymerase (5×)	0.5
Phusion high-fidelity reaction buffer (5×)	10
dNTP mix	1
Ultrapure water	3.5
Total volume	50

34| Perform PCR in a preheated thermocycler, according to the following cycling conditions:

Step	Temp	Time	Cycles
Initial denaturation	98 °C	30 s	1
Denaturation	98 °C	10 s	12–15
Annealing	65 °C	30 s	
Extension	72 °C	30 s	
Final extension	72 °C	5 min	1
	4 °C	Hold	

35| Combine the two MeDIP reactions (do the same for input).

36| Purify the DNA sample with Ampure XP purification beads and elute in 18 µl of EB.
■ **PAUSE POINT** Purified PCR products can be stored at −20 °C for up to 3 months and longer at −80 °C.

37| Add 0.5 µl of PCR-amplified DNA from Step 36 to 0.5 µl of water and run on an Agilent Bioanalyzer 2100 using High-Sensitivity DNA chips and reagents according to the manufacturer's instructions; validate the DNA fragment size range and concentration using the instrument's software.
? TROUBLESHOOTING

Library size selection ● TIMING ~7 h

38| Prepare a 100-ml 2% (wt/vol) TAE LMP agarose gel for each DNA sample (see REAGENT SETUP).

39| Mix the remaining PCR-amplified DNA from Step 36 (after analysis in Step 37) with 3.5 µl of 6× loading dye and load on the gel, leaving space on either side for ladders.
▲ **CRITICAL STEP** Use the recommended loading dye as it is important that the loading dye is heavy enough to prevent sample floating out of the well. Load a 50-bp DNA ladder in the wells flanking the DNA sample. Maintain a minimum of 2 cm (three wells) between the ladder and sample wells.

40| Carry out gel electrophoresis in freshly prepared TAE buffer containing ethidium bromide (0.5 µg ml^{-1}), at 100 V for 100 min.

41| After gel electrophoresis, carefully transfer the gel onto a UV transilluminator.
! CAUTION The gel is very fragile.
▲ **CRITICAL STEP** Place a strip of aluminum foil beneath the lane containing the DNA sample to prevent UV cross-linking of DNA.

42| With a clean scalpel, excise the desired 50-bp library size range. Ensure that the gel slice is cut as close to the DNA smear as is safely possible.
! CAUTION UV exposure can be carcinogenic. Wear protective clothing and UV-resistant face shield.
▲ **CRITICAL STEP** To prevent cross-contamination of libraries, use a fresh scalpel for each library.

43| For contingency purposes, excise one 50-bp size range gel band above and below the library intended for sequencing.

44| Extract and purify the DNA libraries with the MinElute Gel DNA extraction kit and elute in 10 µl of buffer EB. For a detailed description, please see **Box 2**.

45| Assess 1 µl of size-selected DNA on an Agilent Bioanalyzer using High-Sensitivity DNA chips and reagents according to the manufacturer's instructions; determine the concentration and molarity of the excised region with the instrument's software.

46| Dilute 1 μl of size-selected DNA to 10 μl with sterile H_2O and validate the enrichment of genomic regions of known methylation status by qPCR as detailed in Steps 28–30.

47| Calculate enrichment as described in **Box 4**.
▲ CRITICAL STEP Proceed with the next steps only if the MeDIP fraction shows a fold-enrichment ratio of ≥25 for methylated regions in sample DNA.
? TROUBLESHOOTING

Next-generation sequencing ● TIMING ~7 d
48| Subject the sample to next-generation sequencing on the Illumina GAIIx or later models according to the manufacturer's instructions.

Data analysis ● TIMING ~15 h
49| Use FastQC (http://www.bioinformatics.bbsrc.ac.uk/projects/fastqc/) to perform QC of the reads. Check that sequencing data look normal (see FastQC instructions for more information (http://www.bioinformatics.bbsrc.ac.uk/projects/fastqc/Help/3%20Analysis%20Modules/)).

50| Align MeDIP-seq reads to the relevant reference genome using BWA[28].
▲ CRITICAL STEP Other alignment tools such as Bowtie[31] and Novoalign (Novocraft Technologies) may also be used.
▲ CRITICAL STEP Before alignment, BWA should be used to index the reference genome with 'bwa index', using .fasta files as input (the genomic .fasta files are available from numerous public repositories, including the UCSC Genome Browser (http://genome.ucsc.edu/)).

51| Use BWA to run 'bwa aln' on both read ends to generate two .sai output files.

52| Use BWA to run 'bwa sampe' using the .sai files from Step 51 as input; the output is a SAM-formatted alignment file.

53| Use SAMtools[29] to discard reads from the SAM-formatted alignment file that fail to map as a proper pair.

54| Use MeDUSA (available upon request) to discard read pairs in which neither read scored a BWA alignment score of ≥10.

55| Use MeDUSA to identify groups of nonunique reads (i.e., 'clonal reads'—reads that align to the exact same start and stop position on the same chromosome) to discard all but one read from the group.

56| Perform MeDIP-seq–specific QC on the filtered data from Step 55 with MeDIPS[30].
? TROUBLESHOOTING

? TROUBLESHOOTING
Troubleshooting advice can be found in **Table 3**.

TABLE 3 | Troubleshooting table.

Step	Problem	Possible reason	Solution
31	No recovery of MeDIP fraction	IP incubation time is too short	Increase IP incubation time
		Poor or insufficient binding of antibody and/or beads	Check that all reagents have been correctly dispensed at the specified volume and concentration
		Poor construction of control cocktail	Check that control cocktail sequences are methylated using methylation-specific enzyme
			Check that control cocktail sequences amplify independently of MeDIP-seq experiment
			Try using species-specific qPCR primers to validate recovery of MeDIP (expect 10–20% recovery)
		None of the above?	Repeat MeDIP on adapter-ligated DNA

(continued)

TABLE 3 | Troubleshooting table (continued).

Step	Problem	Possible reason	Solution
	Low (< 10%) recovery of MeDIP fraction and low (< 90%) specificity	IP incubation time is too short	Increase IP incubation time
		Aspiration of bead complex following IP (manual only)	If beads remain in tips between MeDIP and purification, pipette-mix the solution until no beads remain in the tip
		None of the above?	Repeat MeDIP on adapter-ligated DNA
	Low (< 10%) recovery but high (>95%) specificity	IP incubation time is too short but MeDIP is not considered to have failed	Proceed to amPCR (Step 41) and evaluate
	C_t values for both QC regions in the MeDIP fraction are similar	IP incubation temperature is too high	Reduce IP incubation temperature to 4 °C
		Incorrect PCR product has been amplified	Check qPCR melt curve
37	No or very low post-amPCR yield	Poor or very inefficient adapter ligation	Check that peak size (assessed with Bioanalyzer) of fragments increases between sonication and adapter ligation
			Dilute post–adapter ligated samples 1 in 10 and perform a single amPCR; you should see a smear on a gel
			Ensure that samples have been purified before amPCR
47	Little or no enrichment of methylated regions in MeDIP fraction	PCR failure	Check the gel for DNA smear
		Gel purification failure	Ensure that adequate volumes of buffer QG are used
			Ensure that pH of QG and sample are correct (< 7.5); if not, add 10 μl of 3 M NaAc (pH 5.0)
		Inaccurate DNA quantification	Use Agilent High-Sensitivity DNA chips. Avoid spectophotometry
		MeDIP and input sample concentrations are not normalized	Normalize MeDIP and Input sample concentrations to >0.1 ng μl^{-1}
		Poor starting DNA quality	Ensure that approximately 25–30% of starting DNA falls into the desired excision range
56	Low proportion of aligned reads	Low-quality base score at 3′ end of read	Trim bases from the sequence before alignment

● **TIMING**

Steps 1–15, DNA/RNA isolation and sonication: ~6 h, day 1
Steps 16–25, library prep A: ~7 h, day 2
Step 26, methylated DNA immunoprecipitation: 23–25 h (usually carried out overnight)
Steps 27–31, qPCR to test for recovery of spiked-in λ-DNA and specificity of the MeDIP reaction (QC1): ~4 h, day 3
Steps 32–37, library preparation B: ~3 h, day 3/4
Steps 38–47, library size selection: ~7 h, day 4/5
Step 48, next-generation sequencing: ~7 d
Steps 49–56, data analysis: ~15 h

ANTICIPATED RESULTS

We anticipate the following results to be achievable now by using this 'nano-MeDIP-seq' protocol:

DNA preparation

Although we show MeDIP-seq to work with as little 50 ng of genomic DNA (diluted from more concentrated stock), we have not tested the protocol on DNA isolated from as few as 10,000–15,000 cells. The minimum we tested was ~75,000 cells, which yields 200–300 ng of genomic DNA. Our expected DNA yield from ~75,000 cells is 200–300 ng (O.T., unpublished data). This depends on the cell type/size, as well as on the method of extraction. Care should be taken to accurately quantify the starting amount of DNA used in MeDIP-seq experiments regardless of the starting number of cells.

Sonication

To minimize DNA loss during library preparation, it is important to use a method resulting in a tight distribution of DNA fragments in the desired size range. By following the above protocol, the Bioruptor sonicator we used consistently generated DNA fragments with a peak at 180–190 bp after approximately 60–90 min of sonication. Reproducibility was highly dependent on the state of the sonicator as well as the temperature during sonication. Other sonicators (e.g., Covaris) and fragmentation methods (e.g., nebulization) can be expected to yield equivalent results but have not been tested here.

Library preparation

At the end of library preparation A (Step 25), DNA fragments should have adapter sequences at both ends, resulting in an increase in size by ~60 bp. We efficiently and consistently used this protocol to enrich for methylated parts of the genome starting with as little as 50 ng with excellent specificity (QC1; O.T., G.A.W., T.M., S.B. and L.M.B., unpublished data). Depending on the starting DNA concentration, 12–15 cycles of amPCR should yield 0.5–3 µg of amplified DNA, which results in a DNA library of 20–150 ng after gel excision. We consistently obtain a minimum enrichment of 25-fold for methylated DNA fragments containing ~10% CpG density in our MeDIP-seq libraries (QC2; O.T., G.A.W., T.M., S.B. and L.M.B., unpublished data).

Sequencing yield

The number of reads depends on the sequencing platform. At the time of writing, an Illumina GAIIx platform should produce between 25 and 35 million reads per lane from the MeDIP-seq libraries generated using this protocol (O.T., G.A.W., T.M., S.B. and L.M.B., unpublished data). Sources of variation in sequencing yield include the following: the sequencing machine itself; protocols used by the sequencing facility; and library quantification before cluster generation.

Bioinformatic analysis

For the majority of the metrics from FastQC, a typical MeDIP-seq sample will generate output that looks consistent with regular genomic DNA sequencing; however, there are some exceptions to be aware of. 'Per sequence GC content' may show deviation from the expected distribution as a result of the enrichment performed on the sample. In addition, it is likely that unusual levels of 'Sequence Duplication' and 'Overrepresented sequences' will be seen, along with enrichment for certain Kmers in the 'Kmer Content' metric. These results are expected in MeDIP-seq because of the enrichment in repetitive DNA.

By using the default alignment parameters, we are able to align in the region of 80–90% of our read pairs. The proportion of reads remaining after filtering will vary on a sample-by-sample and species-by-species basis; however, we would expect to have in the region of 60% of the original reads available for further analysis (although this figure can be lower for mouse samples).

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